

Complete Array (1)

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

An array is called **complete** if it contains at least one positive number, at least one negative number, at least one even number, and at least one odd number.

You may perform the following operation any number of times:

- Add any integer to the array. Each addition costs C .

Determine the minimum total cost required to make the array complete.

Input

The first line contains two integers n and C ($1 \leq n \leq 10$, $1 \leq C \leq 20$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-10 \leq a_i \leq 10$)

Output

Print a single integer: the minimum total cost required.

Examples

standard input	standard output
6 10 -5 -2 3 4 7 -8	0
5 7 2 4 6 -8 10	7